TEXORA · COURSE NOTES · LECTURE SHEET

Periodicity of the Elements

The periodic law and the great repeating trends — atomic and ionic size, ionisation energy, electron affinity and electronegativity — and the reasons underneath them.

CHE-1105 · CHEMISTRY-I · TOPIC 1 · LEVEL 1 TERM 1

1. The periodic law

Chemistry has more than a hundred elements, yet their behaviour is not a hundred separate stories. In 1869 Mendeleev arranged the elements by atomic weight and chemical similarity and found the properties repeating — **periodically** — and he trusted the pattern enough to leave gaps for undiscovered elements and predict their properties. The modern statement, resting on Moseley's X-ray work, is cleaner: **the properties of the elements are a periodic function of their atomic numbers**. The cause is electronic: as atomic number rises, electrons fill shells and subshells in a repeating pattern, and it is the outer (valence) electrons that decide chemistry. Elements in the same vertical **group** share a valence configuration, hence their family resemblance; the horizontal **periods** trace the filling of one shell.

1.1 The map of the table

- **Blocks:** the table divides by the subshell being filled — s-block (Groups 1–2), p-block (Groups 13–18), d-block (transition metals, Groups 3–12), f-block (lanthanides and actinides, placed below).
- **Families with names:** Group 1 alkali metals, Group 2 alkaline earth metals, Group 17 halogens, Group 18 noble gases.
- **The metal line:** metals occupy the left and centre, non-metals the upper right, with the metalloids (B, Si, Ge, As, Sb, Te) strung along the staircase between.

Every trend in this topic is explained by three competing quantities: the **nuclear charge** (Z), the **distance** of the valence electrons from the nucleus (shell number and radius), and **shielding** — the inner electrons' screening of the nuclear pull. The effective pull actually felt is the **effective nuclear charge**, $Z_{eff} \approx Z -$ (shielding). Learn to reason with these three, and no trend needs to be memorised separately.

2. Atomic and ionic radius

Across a period, atomic radius decreases. Electrons are added to the same shell while nuclear charge rises, so Z_{eff} increases and the whole cloud is pulled inward. **Down a group, radius increases:** each step adds a whole new shell, and the extra distance and shielding overwhelm the rise in Z. Ions follow their own logic: a **cation is always smaller than its atom** (an electron is lost, often a whole shell, and the remaining electrons feel a stronger pull per electron); an **anion is always larger** (added electrons repel each other in the same shell). Across an **isoelectronic series** (ions with the same electron count, such as N^{3-}, O^{2-}, F^{-}, Na^{+}, Mg^{2+}, Al^{3+}), radius falls steadily as nuclear charge rises — the same electrons squeezed by more protons.

3. Ionisation energy

The **first ionisation energy** is the energy needed to remove one mole of electrons from one mole of gaseous atoms, forming gaseous ions: $X(g) \rightarrow X^{+}(g) + e^{-}$. It is always endothermic — you pay to pull an electron away from a nucleus. **Across a period it rises** (smaller atoms, higher Z_{eff}, electrons held tighter); **down a group it falls** (valence electrons further out and better shielded). Two small dips interrupt the period rise and both are instructive:

- **Group 2 → Group 13 dip (e.g. Be → B):** the electron removed from B comes from a higher-energy 2p subshell, shielded by the filled 2s pair — slightly easier to remove despite the higher Z.
- **Group 15 → Group 16 dip (e.g. N → O):** nitrogen's 2p subshell is exactly half-filled (one electron in each orbital, stable), while oxygen must pair two electrons in one p orbital; their mutual repulsion makes one easier to remove.

Successive ionisation energies diagnose group number: they climb gradually within a shell, then jump hugely when the removal starts on an inner shell. A jump after the 2nd ionisation ($IE_3 \gg IE_2$) announces a Group 2 element — the third electron must come from a noble-gas core.

> **WORKED EXAMPLE — REASONING WITH THE THREE FACTORS**
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> Arrange Na, Mg and K by first ionisation energy. Na (period 3, group 1): one valence electron in shell 3. Mg: same shell, one more proton — higher Z_{eff}, smaller atom, so IE(Mg) > IE(Na). K: same group as Na but shell 4 — further out, better shielded, so IE(K) < IE(Na). Order: **Mg > Na > K** — the accepted values (≈496, 738 and 419 kJ mol^{-1} for Na, Mg, K respectively) agree.

4. Electron affinity and electronegativity

Electron affinity is the energy change when a gaseous atom gains an electron: $X(g) + e^{-} \rightarrow X^{-}(g)$. It is most exothermic for the halogens (one electron completes the shell) and roughly zero or endothermic for noble gases and half-filled-shell atoms such as nitrogen. Trends are less regular than ionisation

energy because the added electron enters an already negative environment, but the working rule stands: electron affinity becomes more favourable across a period and less favourable down a group, with chlorine — not fluorine — showing the most exothermic value (fluorine's tiny size crowds the incoming electron against the existing ones).

Electronegativity is the power of an atom *in a bond* to attract the shared electrons toward itself (Pauling scale; fluorine is the maximum at 4.0). It rises across a period and falls down a group, mirroring small size and high Z_{eff}. Electronegativity difference predicts bond character: near zero — covalent sharing; moderate — polar covalent; large (roughly above 1.7) — ionic transfer. It is the single most-used number in predicting polarity, and it returns throughout the bonding topic.

5. Further periodic patterns

- **Metallic character** increases down a group and decreases across a period (the mirror of electronegativity): the most metallic element is at the bottom left, the least at the top right.
- **Oxidation states:** for the main groups, the maximum oxidation state commonly equals the group number in the older notation (e.g. +5 for Group 15), and oxides change character across a period from basic (Na_2O) through amphoteric (Al_2O_3) to acidic (SO_3, Cl_2O_7).
- **Diagonal relationships:** the second-period elements resemble the element diagonally below-right — lithium behaves like magnesium, beryllium like aluminium, boron like silicon — because moving right (raising charge density) and moving down (lowering it) cancel.
- **Anomalous second period:** Li, Be, B, C, N, O and F are small, lack d-orbitals, and cap their covalency at 4 — the reason carbon and nitrogen chemistry cannot simply be extrapolated from silicon and phosphorus.

6. Glossary — key terms of this topic

TERM	MEANING
Effective nuclear charge	The nuclear pull actually experienced by an electron after shielding by the others, Z_{eff}.
Shielding	The reduction of nuclear attraction on outer electrons by the inner electrons between them and the nucleus.
Ionisation energy	Energy required to remove an electron from a gaseous atom (first, second, … ionisation).
Electron affinity	Energy change when a gaseous atom gains an electron to form a negative ion.
Electronegativity	A bonded atom's power to attract shared electrons (Pauling scale, F = 4.0).

Isoelectronic	Having the same number and arrangement of electrons (said of ions or atoms).
Amphoteric	Able to react with both acids and bases (e.g. Al_2O_3, ZnO).

7. Key points and summary

- Properties repeat because valence-electron patterns repeat; the whole topic is three factors — nuclear charge, distance, shielding — applied again and again.
- Radius falls across a period and rises down a group; cations shrink, anions swell, and isoelectronic ions shrink as Z rises.
- Ionisation energy rises across and falls down, with the Be → B and N → O dips explained by subshell energy and electron pairing; successive values jump when a new shell is breached.
- Electronegativity (and its mirror, metallic character) summarises an atom's pull on electrons and predicts bond type; diagonal and second-period anomalies complete the map.

8. Practice

Answer in your own words; brief model answers follow.

1. Arrange Li, Be, B and C by atomic radius (largest first) and justify using two of the three master factors.
2. Which is larger, Na or Na^+? Which is larger, Cl or Cl^-? Give the one-line reason for each.
3. In the isoelectronic series O^{2-}, F^-, Na^+, Mg^{2+}, state the order of ionic radius and the factor that decides it.
4. Explain the dip in first ionisation energy from phosphorus to sulphur (the Group 15 → 16 dip).
5. An element's successive ionisation energies (kJ mol^{-1}) are about 590, 1,150, 4,900, 6,500. To which group does it belong, and what in the numbers tells you?
6. Using electronegativity, classify the bonds in NaCl, HCl and Cl_2 as ionic, polar covalent or non-polar covalent.
7. State the diagonal relationship shown by beryllium, and the quantity whose near-equality causes it.
8. Predict the acid–base character of MgO and SO_3 and the trend they belong to.

Answer key

1. Li > Be > B > C — same shell throughout, but nuclear charge rises, so Z_{eff} pulls the cloud progressively inward.
2. Na is larger (Na^+ has lost its whole outer shell). Cl^- is larger (the added electron increases repulsion in the same shell).

3. $O^{2-} > F^{-} > Na^{+} > Mg^{2+}$ — all have 10 electrons; radius falls as nuclear charge rises from 8 to 12.
4. Sulphur’s 3p subshell contains a paired orbital; the repulsion between the paired electrons makes one easier to remove than phosphorus’s half-filled, unpaired arrangement.
5. Group 2 — two moderate values, then a huge jump at the third, which must be torn from an inner (noble-gas) shell.
6. NaCl — ionic (large difference). HCl — polar covalent (moderate difference). Cl_2 — non-polar covalent (zero difference).
7. Beryllium resembles aluminium (diagonally below-right); caused by closely similar charge density (charge-to-size ratio).
8. MgO is basic, SO_3 is acidic — oxides run basic → amphoteric → acidic across a period.
